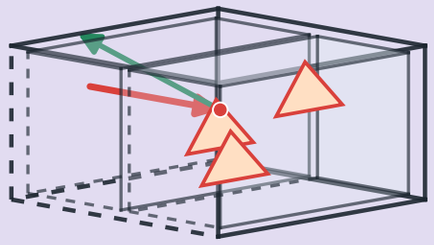
Ray Tracing Attention

Output token

$y_t = \text{argmax}_v \ell_v$

Vocabulary geometry



Token / block identity

Ray Tracing Readout

LM Head geometry

$\ell_v = \sum_b (P_b h)^\top (P_b w_v)$

RMSNorm

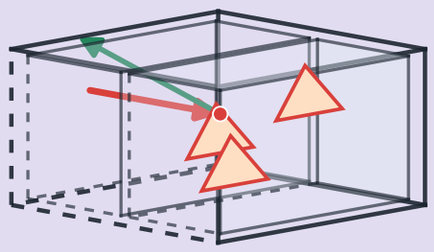
RTA layers

High-D layer state

Restore / next-layer input

$h = P^{-1}(\oplus_b P_b h)$

Weight geometry



Feature BVH / coefficients

$N \times$

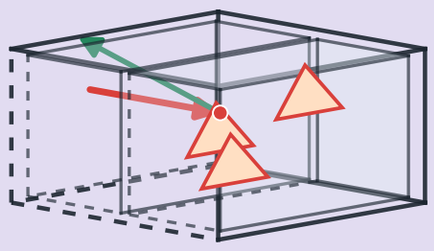
Ray Tracing MLP

Gate / up light response

Down readout / high-D restore

RMSNorm

Shared history BVH



Immutable blocks / live field

Ray Tracing Attention

Projection / RoPE / ray query

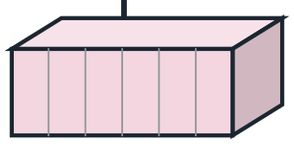
Detector / high-D restore

RMSNorm

Token embedding

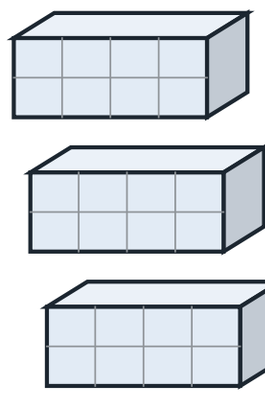
Source model coefficients

$h_0 = \text{Embed}(s_{0:t})$



$s_{0:t}$

Coordinate blocks



$P_0 h$

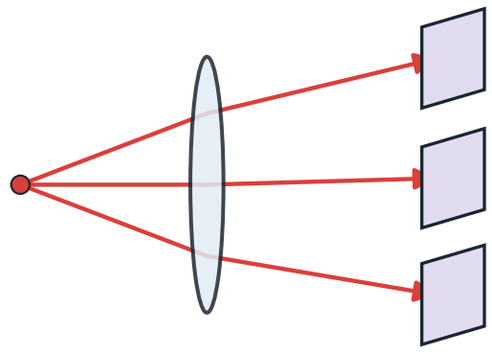
$P_1 h$

$P_b h$

$P^{-1}(\bigoplus_b P_b h) = h$

Complete high-D reconstruction

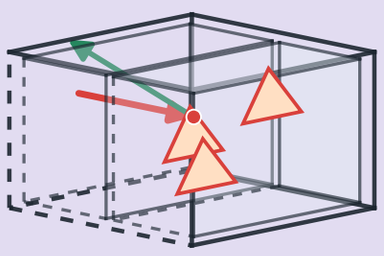
Parallel ray queries



h_ℓ

positions / heads / blocks

Causal address BVH



Token / successor address